

ANANYA RAHAMAN

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with 3+ years of experience in designing and developing scalable full-stack applications. Proficient in Java, Python, JavaScript, and SQL, with hands-on expertise in React, Node.js, Express, and Spring Boot. Experienced in building RESTful APIs, microservices, and cloud-based solutions on AWS, Azure, and GCP. Strong background in CI/CD, Docker, Kubernetes, and Agile/Scrum development practices. Passionate about writing clean, efficient code and delivering robust, high-performance software systems.

SKILLS

- **Programming Languages:** Python, R, SQL, Java, C++, JavaScript, Bash, MATLAB.
- **Databases:** SQL, NoSQL, MySQL, PostgreSQL, MongoDB, Redis.
- **Frontend Development:** HTML, CSS, React.js, vue.js, Next.js, Redux, TailwindCSS, Webpack, Responsive UI, Cross-Browser Compatibility, UI performance Optimization.
- **Backend Development and Frameworks:** Node.js, Express.js, Microservices, REST APIs, SOAP, Spring Boot, Junit, Spring Cloud, Hibernate, Laravel, Event-Driven Architecture.
- **DevOps & Tools:** CI/CD, Jenkins, GitHub Action, Docker, Kubernetes, Git, GitHub, GitLab, Code Reviews, Debugging, Data Visualization (Plotly, Power BI), Testing (Jest, PyTest).
- **Cloud Platform and Project Management:** AWS, GCP, Microsoft Azure, Agile & SDLC, Scrum, Jira, Slack.
- **AI and ML:** LLMs, Deep Learning, Computer Vision, OpenCV, NLP, Data Augmentation & Synthetic Data Generation.
- **Hardware:** IoT, Arduino, GSM Module, RaspberryPi.
- **Others:** Json, XML

WORK EXPERIENCE

Graduate Research and Teaching Assistant, 09/2023 - 04/2025

University of Western Ontario – London, ON, Canada

- Guided students in **full-stack development** projects using **React, Node.js, Express, and SQL**, reinforcing best practices in modular design and version control (Git/GitHub).
- Reviewed and debugged student code, ensuring adherence to **clean code principles**, scalability, and maintainability standards.
- Conducted tutorials and lab sessions on **software development lifecycles (SDLC)**, **API integration**, and **object-oriented programming (OOP)** in **Java and Python**.
- Mentored students in **team-based software projects**, emphasizing **Agile/Scrum**, **version control workflows**, and **collaborative development practices**.

Lecturer, Computer Science, 02/2021 - 08/2023

Primeasia University – Dhaka, Bangladesh

- Designed and delivered advanced undergraduate and graduate courses in **Software Engineering**, **system optimization**, **System Design**, and **Applied AI Systems**, focusing on **scalable architectures** and **enterprise-grade software practices**.
- Created comprehensive **curricula and instructional materials**, introducing students to **cloud-native design**, **DevOps workflows**, and **microservices architecture**.
- Supervised student **theses and independent research projects**, guiding them in developing **real-world prototypes** using **React, Node.js, Docker**, and **AWS**.

- Delivered lectures on **software reliability, metrics, and large-scale system analysis**, and organized industry-academia **workshops** to connect theory with modern software practices.

Software Engineer, 01/2019 - 10/2020

Therap BD (LTD) – Dhaka, Bangladesh

- Designed and developed **scalable backend systems and RESTful APIs** using **Python, Java, Node.js, and SQL**, integrating with cloud services (**AWS/GCP/Azure**) for deployment.
- Built **responsive and interactive web applications** with **React, JavaScript/TypeScript, and modern UI frameworks**, ensuring high performance and user experience.
- Implemented **microservices architectures** with containerization tools (**Docker, Kubernetes**) to enhance scalability and reliability.
- Engineered **data pipelines** (ETL workflows) for processing large structured and unstructured datasets, leveraging **SQL, NoSQL (MongoDB), and PostgreSQL/MySQL/Oracle** databases.
- Integrated **machine learning models** into production systems using **TensorFlow, PyTorch, and scikit-learn**, enabling intelligent features in applications.
- Applied **CI/CD pipelines** (Jenkins, GitHub Actions, GitLab CI) to automate testing, integration, and deployment.
- Built **data visualization dashboards** using **Plotly, D3.js, and Power BI** to monitor KPIs, system health, and ML performance.
- Collaborated in **Agile/Scrum teams**, contributing to sprint planning, peer code reviews, and cross-functional development.

EDUCATION

Master of Science: Computer Science, GPA-3.93

04/2025

University of Western Ontario - London, ON

Bachelor of Science: Computer Science And Engineering, GPA-3.92

10/2020

BRAC University - Dhaka, Bangladesh

CERTIFICATIONS

- Introduction to Software Engineering [Coursera]
- IBM Devops and Software Engineering [Coursera]
- IBM Full stack Software Developer [Coursera]

PUBLICATIONS

Evaluating SQL Understanding in Large Language Models

EDBT2025 [paper's link](#)

An IOT Based Smart Drain Monitoring System with Alert Messages

IHCI2020 [paper's link](#)

Design and Implementation of a safety device for emergency situation using Arduino and GSM Module

IHCI2020 [paper's link](#)

PROJECTS

- **Online Tutor-Student Platform:** Built a web platform that connects tutors and students based on subject preferences and availability. Implemented profile management, search filters, and chat functionality using **Django + React**. Integrated **Stripe API** for secure payments.
- **Liability & Asset Workflow:** An automated workflow and inventory management system for a commercial bank, allowing the integration of multiple banking software (i.e., Core Banking, Agent Banking, Card Management System). Technology used: Java, Spring Boot, Oracle DB, ReactJS, Jenkins, Apache Tomcat, Nginx.